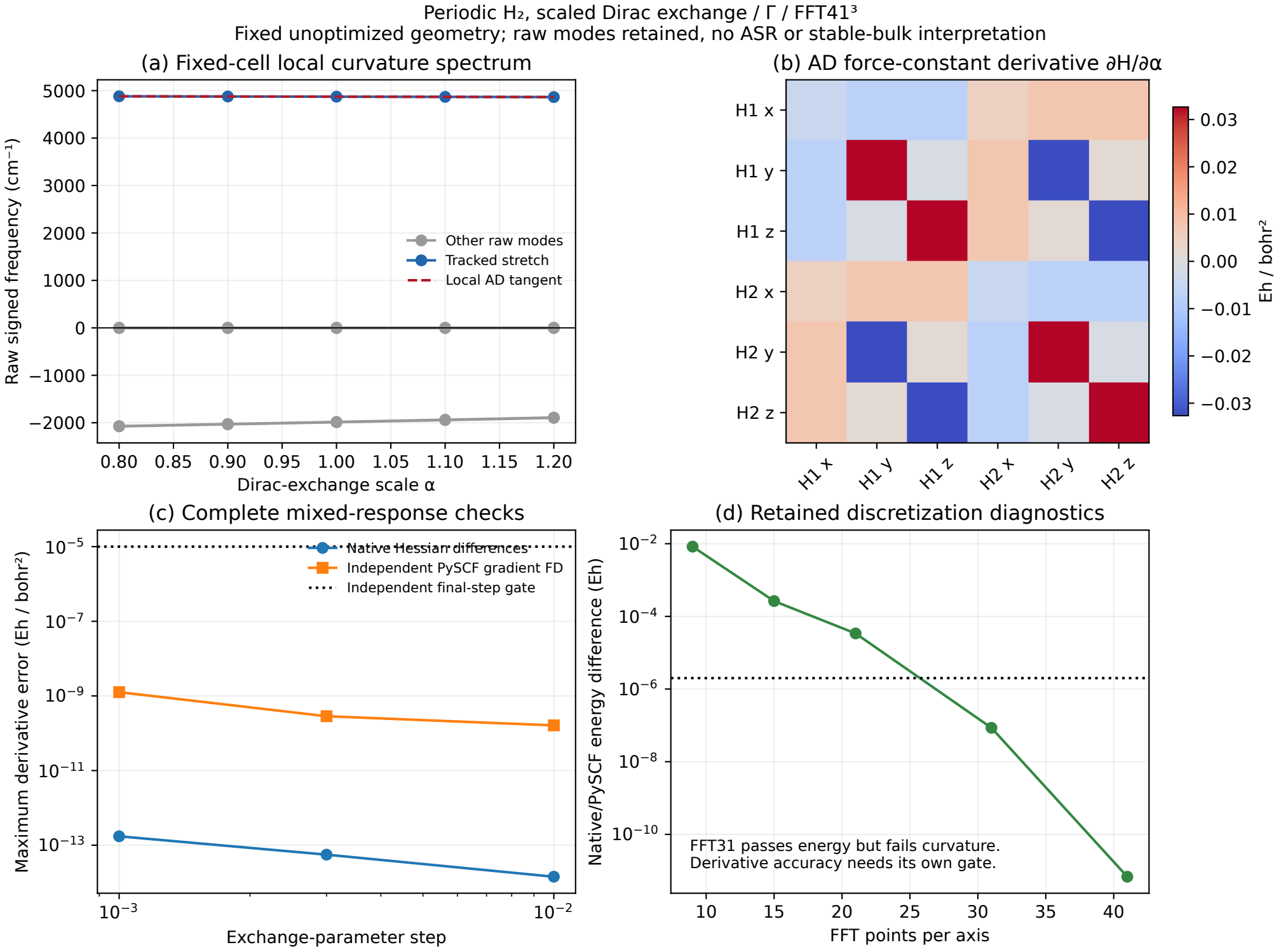
Periodic H₂, scaled Dirac exchange / Γ / FFT41³
Fixed unoptimized geometry; raw modes retained, no ASR or stable-bulk interpretation
(a) Fixed-cell local curvature spectrum
Raw signed frequency (cm⁻¹)
Dirac-exchange scale α
Other raw modes
Tracked stretch
Local AD tangent
(b) AD force-constant derivative ∂H/∂α
H1 x
H1 y
H1 z
H2 x
H2 y
H2 z
Eh / bohr²
(c) Complete mixed-response checks
Maximum derivative error (Eh / bohr²)
Native Hessian differences
Independent PySCF gradient FD
Independent final-step gate
Exchange-parameter step
(d) Retained discretization diagnostics
Native/PySCF energy difference (Eh)
FFT31 passes energy but fails curvature.
Derivative accuracy needs its own gate.
FFT points per axis